

Generation Interconnection System Impact Study Report

For Queue Project Q20-1422

Cisco Grove 345 kV

June 01, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1422. The Interconnection Customer proposes a 80.5 MW solar photovoltaic facility interconnecting at the Cisco Grove 345 kV substation in the Pennsylvania area. The request was received on March 10, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 496 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$120,250,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Dwp – Humacao Tc 138 KV line	Chambers Springs – Jh Warden 500 KV	177.6	195	5.3
Humacao Tc – Strasburg Substation 138 KV line	Jh Warden – Chambers Springs 500 KV	170.6	205	5.3
Strasburg – Perryton 138 KV	Chambers Springs – Jh Warden 500 KV line	169.6	208	5.3
Lamar – Perryton 138 KV	Chambers Springs – Jh Warden Substation 500 KV line	162.3	222	5.3
Lawton E Side – Lamar 138 KV line	Jh Warden – Chambers Springs 500 KV L	158.3	233	5.3
Stony Pt – Cypress Gardens 138 KV	Jh Warden – Chambers Springs Substation 500 KV	141.0	218	5.1
Stony Pt – Panorama 138 KV line	Chambers Springs – J Warden 500 KV	138.0	227	5.1
Chambers Springs 500/345 kV TX	Chambers Springs – Archbald Power Station 500 KV L	137.6	766	13.0
Highway Jct 1 – Panorama Substation 138 KV	Jh Warden – Chambers Springs 500 KV	137.0	234	5.1
San Simeon-Stefano Substation – Highway Jct 1 138 KV	W Temple 345-138 kV Auto	109.4	231	5.8
Puntledge – San Simeon-Stefano 138 KV line	Jh Warden – Cisco Grove 345 KV line	106.5	159	16.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dwp – Humacao Tc 138 KV line	Rebuild line with higher-capacity conductor	\$21,889,000	100% (\$21,889,000)
Strasburg – Humacao Tc 138 KV	Rebuild line with higher-capacity conductor	\$20,036,000	100% (\$20,036,000)
Perryton – Strasburg 138 KV	Rebuild line with higher-capacity conductor	\$6,549,000	100% (\$6,549,000)
Lamar – Perryton 138 KV	Rebuild line with higher-capacity conductor	\$4,646,000	100% (\$4,646,000)
Lawton E Side – Lamar 138 KV line	Rebuild line with higher-capacity conductor	\$9,421,000	100% (\$9,421,000)
Stony Pt – Cypress Gardens 138 KV	Rebuild line with higher-capacity conductor	\$19,789,000	100% (\$19,789,000)
Panorama Substation – Stony Point 138 KV	Rebuild line with higher-capacity conductor	\$3,138,000	100% (\$3,138,000)
Chambers Springs 500-345 kV XFMR	Replace transformer with higher-rated unit	\$15,849,000	100% (\$15,849,000)
Highway Junction 1 – Panorama 138 KV	Rebuild line with higher-capacity conductor	\$15,898,000	100% (\$15,898,000)
San Simeon-Stefano Substation – Highway Jct 1 138 KV	Reconductor line	\$1,746,000	100% (\$1,746,000)
Puntledge – San Simeon-Stefano 138 KV line	Reconductor line	\$1,289,000	100% (\$1,289,000)
	Total allocated to project		\$120,250,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.